In [1]:
```
import numpy as np
import pandas as pd
from matplotlib.pyplot import subplots
```

In [3]:
```
import statsmodels.api as sm
```

In [5]:
```
from statsmodels.stats.outliers_influence import variance_inflation_factor as VIF
from statsmodels.stats.anova import anova_lm
```

In [7]:
```
from ISLP import load_data
from ISLP.models import (
    ModelSpec as MS,
    summarize,
    poly
)
```

In [9]:
```
dir()
```

Out[9]:
```
['In',
 'MS',
 'Out',
 'VIF',
 '_',
 '_2',
 '_4',
 '_6',
 '_8',
 '__',
 '___',
 '__builtin__',
 '__builtins__',
 '__doc__',
 '__loader__',
 '__name__',
 '__package__',
 '__pandas',
 '__session__',
 '__spec__',
 '_dh',
 '_i',
 '_i1',
 '_i2',
 '_i3',
 '_i4',
 '_i5',
 '_i6',
 '_i7',
 '_i8',
 '_i9',
 '_ih',
 '_ii',
 '_iii',
 '_oh',
 'anova_lm',
 'dataframe_columns',
 'dataframe_hash',
 'dtypes_str',
 'exit',
 'get_dataframes',
 'get_ipython',
 'getpass',
 'hashlib',
 'import_pandas_safely',
 'is_data_frame',
 'json',
 'load_data',
 'np',
 'open',
 'pd',
 'poly',
 'quit',
 'sm',
 'subplots',
 'summarize']
```

In [11]:
```
A = np.array([3,5,11])
dir(A)
```

Out[11]:
```
['T',
 '__abs__',
 '__add__',
 '__and__',
 '__array__',
 '__array_finalize__',
 '__array_function__',
 '__array_interface__',
 '__array_prepare__',
 '__array_priority__',
 '__array_struct__',
 '__array_ufunc__',
 '__array_wrap__',
 '__bool__',
 '__buffer__',
 '__class__',
 '__class_getitem__',
 '__complex__',
 '__contains__',
 '__copy__',
 '__deepcopy__',
 '__delattr__',
 '__delitem__',
 '__dir__',
 '__divmod__',
 '__dlpack__',
 '__dlpack_device__',
 '__doc__',
 '__eq__',
 '__float__',
 '__floordiv__',
 '__format__',
 '__ge__',
 '__getattribute__',
 '__getitem__',
 '__getstate__',
 '__gt__',
 '__hash__',
 '__iadd__',
 '__iand__',
 '__ifloordiv__',
 '__ilshift__',
 '__imatmul__',
 '__imod__',
 '__imul__',
 '__index__',
 '__init__',
 '__init_subclass__',
 '__int__',
 '__invert__',
 '__ior__',
 '__ipow__',
 '__irshift__',
 '__isub__',
 '__iter__',
 '__itruediv__',
 '__ixor__',
 '__le__',
 '__len__',
 '__lshift__',
```

```
'__lt__',
'__matmul__',
'__mod__',
'__mul__',
'__ne__',
'__neg__',
'__new__',
'__or__',
'__pos__',
'__pow__',
'__radd__',
'__rand__',
'__rdivmod__',
'__reduce__',
'__reduce_ex__',
'__repr__',
'__rfloordiv__',
'__rlshift__',
'__rmatmul__',
'__rmod__',
'__rmul__',
'__ror__',
'__rpow__',
'__rrshift__',
'__rshift__',
'__rsub__',
'__rtruediv__',
'__rxor__',
'__setattr__',
'__setitem__',
'__setstate__',
'__sizeof__',
'__str__',
'__sub__',
'__subclasshook__',
'__truediv__',
'__xor__',
'all',
'any',
'argmax',
'argmin',
'argpartition',
'argsort',
'astype',
'base',
'byteswap',
'choose',
'clip',
'compress',
'conj',
'conjugate',
'copy',
'ctypes',
'cumprod',
'cumsum',
'data',
'diagonal',
'dot',
'dtype',
'dump',
```

```
 'dumps',
 'fill',
 'flags',
 'flat',
 'flatten',
 'getfield',
 'imag',
 'item',
 'itemset',
 'itemsize',
 'max',
 'mean',
 'min',
 'nbytes',
 'ndim',
 'newbyteorder',
 'nonzero',
 'partition',
 'prod',
 'ptp',
 'put',
 'ravel',
 'real',
 'repeat',
 'reshape',
 'resize',
 'round',
 'searchsorted',
 'setfield',
 'setflags',
 'shape',
 'size',
 'sort',
 'squeeze',
 'std',
 'strides',
 'sum',
 'swapaxes',
 'take',
 'tobytes',
 'tofile',
 'tolist',
 'tostring',
 'trace',
 'transpose',
 'var',
 'view']
```

In [13]:
```
A.sum()
```

Out[13]:
```
19
```

In [15]:
```
Boston = load_data("Boston")
Boston.columns
```

Out[15]:
```
Index(['crim', 'zn', 'indus', 'chas', 'nox', 'rm', 'age', 'dis', 'rad', 'tax
       'ptratio', 'lstat', 'medv'],
      dtype='object')
```

In [17]:
```
X = pd.DataFrame({
      'intercept': np.ones(Boston.shape[0]),
      'lstat': Boston['lstat']
})

X[:4]
```

Out[17]:

	intercept	lstat
0	1.0	4.98
1	1.0	9.14
2	1.0	4.03
3	1.0	2.94

In [19]:
```
y = Boston['medv']
model = sm.OLS(y, X)
results = model.fit()
```

In [21]:
```
summarize(results)
```

Out[21]:

	coef	std err	t	P>|t|
intercept	34.5538	0.563	61.415	0.0
lstat	-0.9500	0.039	-24.528	0.0

In [23]:
```
design = MS(['lstat'])
design = design.fit(Boston)
X = design.transform(Boston)
X[:4]
```

Out[23]:

	intercept	lstat
0	1.0	4.98
1	1.0	9.14
2	1.0	4.03
3	1.0	2.94

In [25]:
```
#fit_transform() method#
design = MS(['lstat'])
X = design.fit_transform(Boston)

X[:4]
```

Out[25]:

	intercept	lstat
0	1.0	4.98
1	1.0	9.14
2	1.0	4.03
3	1.0	2.94

In [27]:
```
results.summary()
```

Out[27]: OLS Regression Results

Dep. Variable:	medv	**R-squared:**	0.544
Model:	OLS	**Adj. R-squared:**	0.543
Method:	Least Squares	**F-statistic:**	601.6
Date:	Mon, 17 Aug 2026	**Prob (F-statistic):**	5.08e-88
Time:	09:24:02	**Log-Likelihood:**	-1641.5
No. Observations:	506	**AIC:**	3287.
Df Residuals:	504	**BIC:**	3295.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
intercept	34.5538	0.563	61.415	0.000	33.448	35.659
lstat	-0.9500	0.039	-24.528	0.000	-1.026	-0.874

Omnibus:	137.043	**Durbin-Watson:**	0.892
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	291.373
Skew:	1.453	**Prob(JB):**	5.36e-64
Kurtosis:	5.319	**Cond. No.**	29.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly speci

In [29]:
```
results.params
```

Out[29]:
```
intercept    34.553841
lstat        -0.950049
dtype: float64
```

In [31]:
```
#The get_prediction()#
new_df = pd.DataFrame({'lstat':[5, 10, 15]})
newX = design.transform(new_df)
newX
```

Out[31]:

	intercept	lstat
0	1.0	5
1	1.0	10
2	1.0	15

In [33]:
```
new_predictions = results.get_prediction(newX);
new_predictions.predicted_mean
```

Out[33]:
```
array([29.80359411, 25.05334734, 20.30310057])
```

In [35]:
```
new_predictions.conf_int(alpha=0.05)
```

Out[35]:
```
array([[29.00741194, 30.59977628],
       [24.47413202, 25.63256267],
       [19.73158815, 20.87461299]])
```

In [37]:
```
#Prediction intervals#
new_predictions.conf_int(obs=True , alpha=0.05)
```

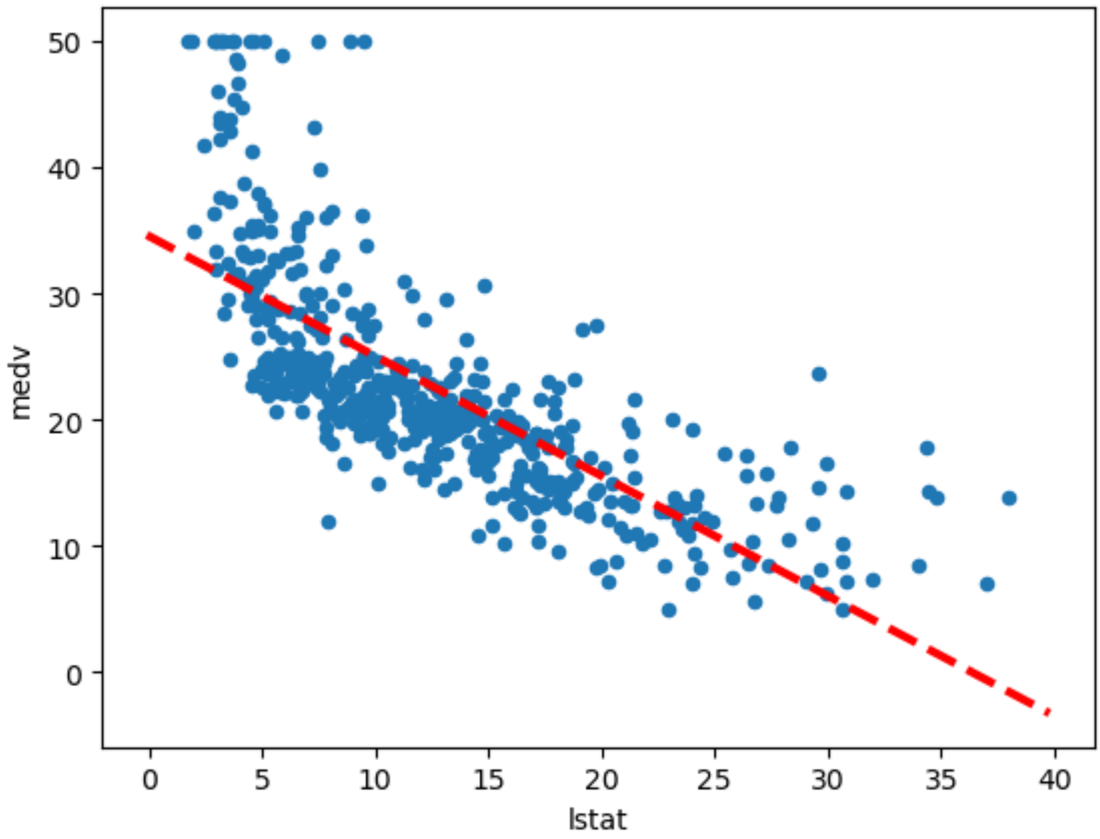
Out[37]:
```
array([[17.56567478, 42.04151344],
       [12.82762635, 37.27906833],
       [ 8.0777421 , 32.52845905]])
```

In [39]:
```
#Defining Functions#
def abline(ax, b, m):
    """Add a line with slope m and intercept b to ax."""
      xlim = ax.get_xlim()
      ylim = [m * xlim[0] + b, m * xlim[1] + b]
      ax.plot(xlim, ylim)
```

In [41]:
```
def abline(ax, b, m, *args, **kwargs):
    """Add a line with slope m and intercept b to ax."""
      xlim = ax.get_xlim()
      ylim = [m * xlim[0] + b, m * xlim[1] + b]
      ax.plot(xlim, ylim, *args, **kwargs)
```

In [49]:
```
ax = Boston.plot.scatter('lstat', 'medv')

abline(
      ax,
      results.params.iloc[0],
      results.params.iloc[1],
      'r--',
      linewidth=3
)
```



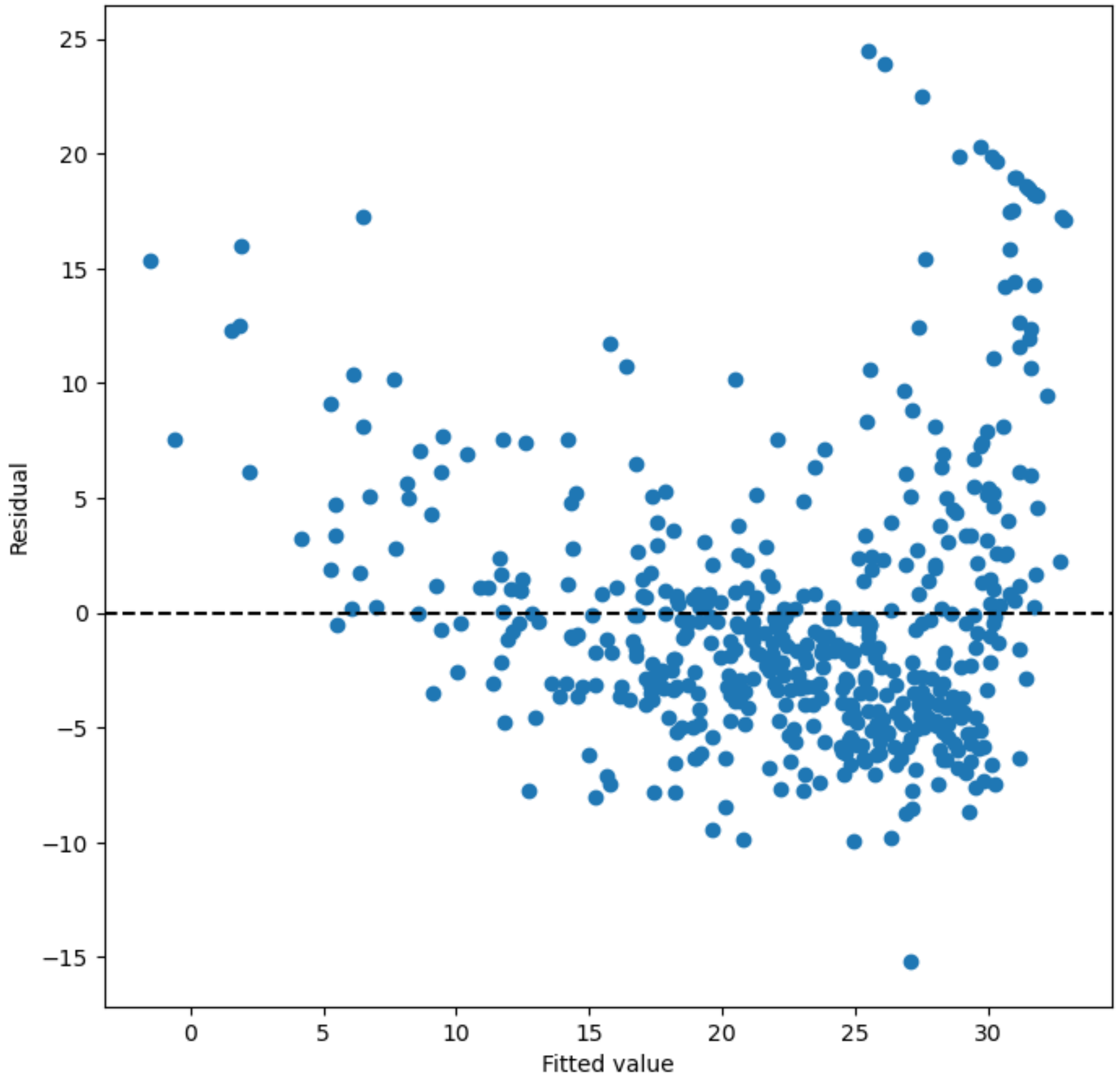
In [53]:

```
            #Linear Regression#
ax = subplots(figsize=(8, 8))[1]

ax.scatter(results.fittedvalues, results.resid)
ax.set_xlabel('Fitted value')
ax.set_ylabel('Residual')
ax.axhline(0, c='k', ls='--')
```

Out[53]:

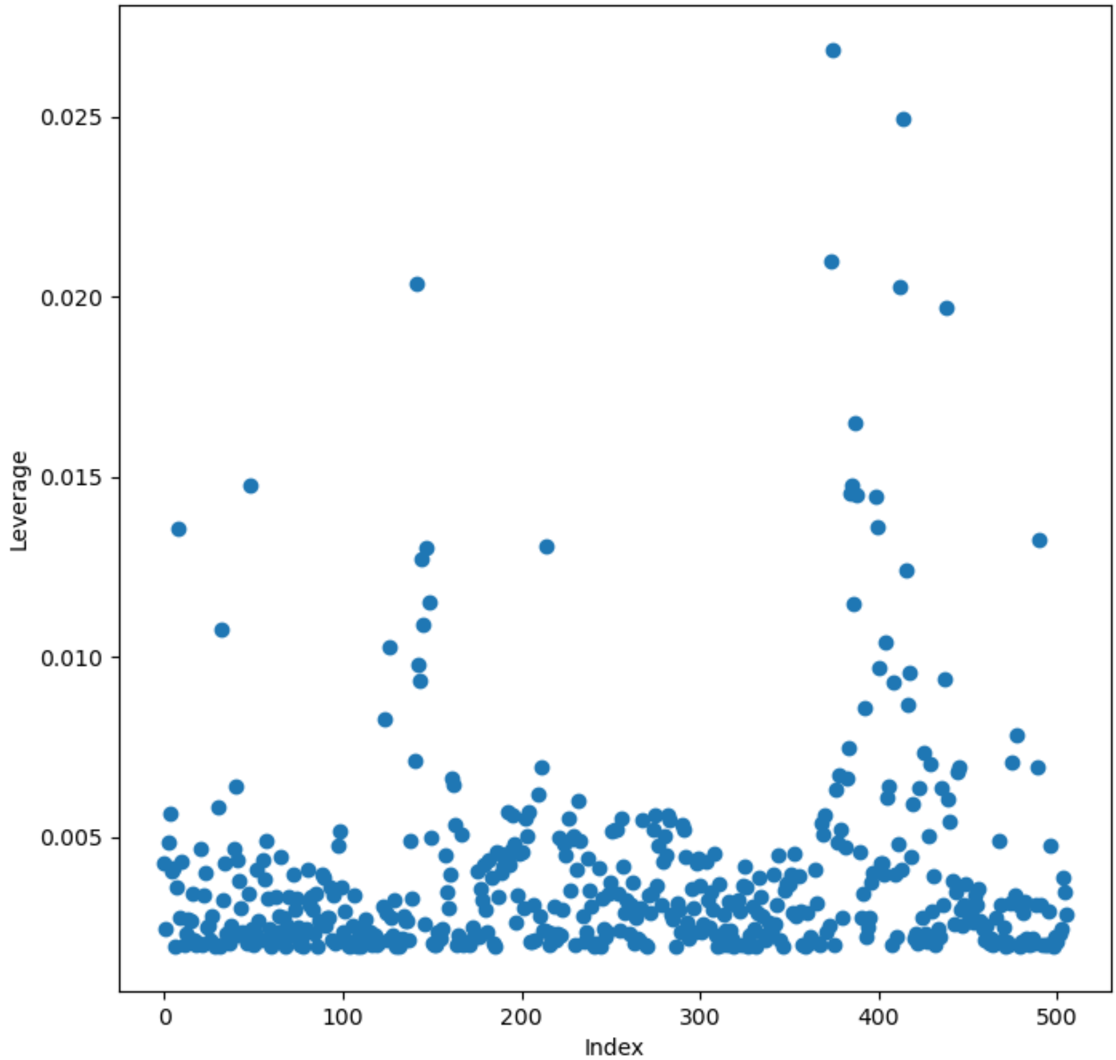
```
<matplotlib.lines.Line2D at 0x1cbaaed1340>
```



In [55]:

```
infl = results.get_influence()
ax = subplots(figsize=(8, 8))[1]
ax.scatter(
        np.arange(X.shape[0]),
        infl.hat_matrix_diag
)
ax.set_xlabel('Index')
ax.set_ylabel('Leverage')
np.argmax(infl.hat_matrix_diag)
```

Out[55]: 374



In [57]:
```
#Multiple Linear Regression#
X = MS(['lstat', 'age']).fit_transform(Boston)
model1 = sm.OLS(y, X)
results1 = model1.fit()
summarize(results1)
```

Out[57]:

	coef	std err	t	P>|t|
intercept	33.2228	0.731	45.458	0.000
lstat	-1.0321	0.048	-21.416	0.000
age	0.0345	0.012	2.826	0.005

In [63]:
```
terms = Boston.columns.drop('medv')
```

In [65]:
```
Boston.columns
```

Out[65]:
```
Index(['crim', 'zn', 'indus', 'chas', 'nox', 'rm', 'age', 'dis', 'rad', 'ta
       'ptratio', 'lstat', 'medv'],
      dtype='object')
```

In [67]:
```
X = MS(terms).fit_transform(Boston)
model = sm.OLS(y, X)
results = model.fit()
summarize(results)
```

Out[67]:

	coef	std err	t	P>|t|
intercept	41.6173	4.936	8.431	0.000
crim	-0.1214	0.033	-3.678	0.000
zn	0.0470	0.014	3.384	0.001
indus	0.0135	0.062	0.217	0.829
chas	2.8400	0.870	3.264	0.001
nox	-18.7580	3.851	-4.870	0.000
rm	3.6581	0.420	8.705	0.000
age	0.0036	0.013	0.271	0.787
dis	-1.4908	0.202	-7.394	0.000
rad	0.2894	0.067	4.325	0.000
tax	-0.0127	0.004	-3.337	0.001
ptratio	-0.9375	0.132	-7.091	0.000
lstat	-0.5520	0.051	-10.897	0.000

In [69]:

```
minus_age = Boston.columns.drop(['medv', 'age'])
Xma = MS(minus_age).fit_transform(Boston)
model1 = sm.OLS(y, Xma)
summarize(model1.fit())
```

Out[69]:

	coef	std err	t	P>|t|
intercept	41.5251	4.920	8.441	0.000
crim	-0.1214	0.033	-3.683	0.000
zn	0.0465	0.014	3.379	0.001
indus	0.0135	0.062	0.217	0.829
chas	2.8528	0.868	3.287	0.001
nox	-18.4851	3.714	-4.978	0.000
rm	3.6811	0.411	8.951	0.000
dis	-1.5068	0.193	-7.825	0.000
rad	0.2879	0.067	4.322	0.000
tax	-0.0127	0.004	-3.333	0.001
ptratio	-0.9346	0.132	-7.099	0.000
lstat	-0.5474	0.048	-11.483	0.000

In [71]:

```
#Multivariate Goodness of Fit#
#List Comprehension#
vals = [VIF(X, i)
        for i in range(1, X.shape[1])]

vif = pd.DataFrame(
    {'vif': vals},
```

```python
    index=X.columns[1:]
)

vif
```

Out[71]:

	vif
crim	1.767486
zn	2.298459
indus	3.987181
chas	1.071168
nox	4.369093
rm	1.912532
age	3.088232
dis	3.954037
rad	7.445301
tax	9.002158
ptratio	1.797060
lstat	2.870777

In [73]:

```python
vals = []

for i in range(1, X.values.shape[1]):
    vals.append(VIF(X.values, i))
```

In [75]:

```python
#Interaction Terms#
X = MS([
    'lstat',
    'age',
    ('lstat', 'age')
]).fit_transform(Boston)
model2 = sm.OLS(y, X)
summarize(model2.fit())
```

Out[75]:

	coef	std err	t	P>|t|
intercept	36.0885	1.470	24.553	0.000
lstat	-1.3921	0.167	-8.313	0.000
age	-0.0007	0.020	-0.036	0.971
lstat:age	0.0042	0.002	2.244	0.025

In [77]:

```python
#Non-linear Transformations of the Predictors#
X = MS([
    poly('lstat', degree=2),
    'age'
]).fit_transform(Boston)
model3 = sm.OLS(y, X)
results3 = model3.fit()
summarize(results3)
```

Out[77]:

	coef	std err	t	P>|t|
intercept	17.7151	0.781	22.681	0.0
poly(lstat, degree=2)[0]	-179.2279	6.733	-26.620	0.0
poly(lstat, degree=2)[1]	72.9908	5.482	13.315	0.0
age	0.0703	0.011	6.471	0.0

In [79]:

```
anova_lm(results1 , results3)
```

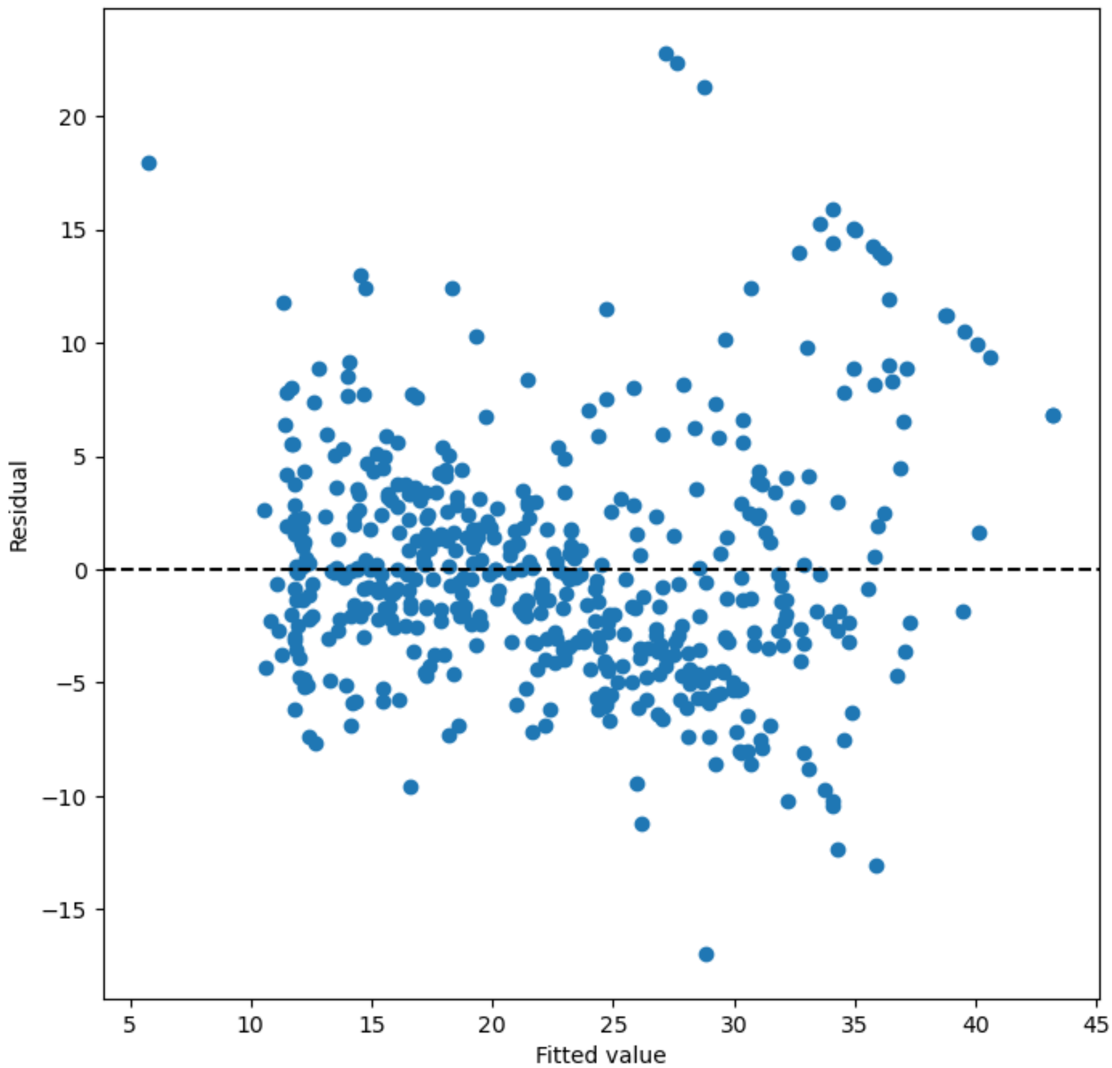
Out[79]:

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	503.0	19168.128609	0.0	NaN	NaN	NaN
1	502.0	14165.613251	1.0	5002.515357	177.278785	7.468491e-35

In [81]:

```
#Linear Regression#
ax = subplots(figsize=(8, 8))[1]
ax.scatter(results3.fittedvalues, results3.resid)
ax.set_xlabel('Fitted value')
ax.set_ylabel('Residual')
ax.axhline(0, c='k', ls='--')
```

Out[81]: <matplotlib.lines.Line2D at 0x1cbab0b8260>



In [83]:
```python
#Qualitative Predictors#
Carseats = load_data('Carseats')
Carseats.columns
```

Out[83]:
```
Index(['Sales', 'CompPrice', 'Income', 'Advertising', 'Population', 'Price',
       'ShelveLoc', 'Age', 'Education', 'Urban', 'US'],
      dtype='object')
```

In [85]:
```python
allvars = list(Carseats.columns.drop('Sales'))

y = Carseats['Sales']
final = allvars + [
    ('Income', 'Advertising'),
    ('Price', 'Age')
]
X = MS(final).fit_transform(Carseats)
model = sm.OLS(y, X)
summarize(model.fit())
```

Out[85]:

	coef	std err	t	P>|t|
intercept	6.5756	1.009	6.519	0.000
CompPrice	0.0929	0.004	22.567	0.000
Income	0.0109	0.003	4.183	0.000
Advertising	0.0702	0.023	3.107	0.002
Population	0.0002	0.000	0.433	0.665
Price	-0.1008	0.007	-13.549	0.000
ShelveLoc[Good]	4.8487	0.153	31.724	0.000
ShelveLoc[Medium]	1.9533	0.126	15.531	0.000
Age	-0.0579	0.016	-3.633	0.000
Education	-0.0209	0.020	-1.063	0.288
Urban[Yes]	0.1402	0.112	1.247	0.213
US[Yes]	-0.1576	0.149	-1.058	0.291
Income:Advertising	0.0008	0.000	2.698	0.007
Price:Age	0.0001	0.000	0.801	0.424

In [ ]: